

Shirou Jing

sj186@rice.edu • Homepage • GitHub • LinkedIn

RESEARCH INTERESTS

Generative models; efficient (diffusion) language-model pretraining; machine learning; image and video generation. I aim to use computer science and AI to create valuable innovations for society.

EDUCATION

Rice University, Houston, TX, USA

Aug 2025 – Present

Ph.D. in Electrical & Computer Engineering. Advisor: Prof. Tong (Tony) Geng.

University of Chinese Academy of Sciences (UCAS), Beijing, China

Sep 2021 – Jun 2025

B.S. in Computer Science. GPA: 3.87/4.0.

Selected courses: Probability & Statistics (4.0), Data Structures (4.0), Theory of Computation (3.9), Human-Computer Interaction (4.0), Digital Circuits (4.0).

University of California, Berkeley, CA, USA

Jan 2024 – May 2024

Exchange Student, Computer Science. GPA: 3.9/4.0.

Selected courses: CS 188 Introduction to AI (4.0); CS 170 Efficient Algorithms & Intractable Problems (top 3%, 4.0).

PUBLICATIONS

- **Shirou Jing**, Chunshu Wu, Chuan Liu, Arghavan Bahadorinejad, Feitong Qiao, Dongfang Liu, Tong Geng. **InfoDLM: an Information-Adaptive Framework for Discrete Diffusion Language Model Pretraining**. *ICML 2026*.
- **Shirou Jing**^{*}, Xuejin Wu, Wei Sun. **GIGT: Gesture-Recognition-Integrated Game Therapy for Tenosynovitis**. *IPCV 2024* (acceptance rate 19–21%).
- Kang Liu, Zuzheng Kuang, **Shirou Jing**, Haixia Bi^{*}, Yaochen Li, Chen Xu. **Diffusion-based Generative Self-supervised Model for Few-shot PolSAR Image Classification**. *IGARSS 2024*.
- Ningwei Wang, Weiqiang Jin, **Shirou Jing**, Haixia Bi^{*}, Guang Yang. **Learning with Noisy Labels via Mamba and Entropy KNN Framework**. *Applied Soft Computing*, 2025.

^{*} Lead / corresponding author where marked.

PREPRINTS & WORK IN SUBMISSION

- **RCS: Rhythm-Controlled Communication for Multi-Agent LLM Systems**. *Submitted to NeurIPS 2026*.
- **CtrLoRA-XL: Cost-efficient, High-fidelity Multi-condition Controllable Image Generation on Stable Diffusion XL**. With Dr. Zhenliang He and Prof. Shiguang Shan. *Submitted to IJCV*.

RESEARCH EXPERIENCE

Graduate Research Assistant — Rice University (Geng Lab)

Aug 2025 – Present

Advisor: Prof. Tong (Tony) Geng. *Information-efficient pretraining of language models; multi-agent systems*.

- **InfoDLM** (ICML 2026): reformulated the heuristic random masking in diffusion-language-model pretraining as an active, information-gain-driven mask policy — a trainable information-gain signal plus maturity-aware masking, jointly trained with the model — improving reasoning accuracy by up to 13% over LLaDA-472M at about 7.6% FLOPs overhead.
- **InfoLLM** (in progress): designing *dynamic, model-maturity-aware* learning tasks so that LLMs begin to acquire reasoning capability from the pretraining stage.

- **RCS** (multi-agent systems): reformulated communication in multi-agent LLM systems from fixed, top-down turn-taking into a learned, self-adaptive *communication rhythm* — coordinating *when* agents contribute, not just *what* — reaching debate-level accuracy at a fraction of the communication cost. Submitted to NeurIPS 2026.

AI-driven High-resolution Cell Dynamic Imaging

Apr 2024 – Oct 2024

Advanced Bioimaging Center, UC Berkeley (co-founded by Nobel laureate Eric Betzig). Research Assistant. Advisors: Dr. Xiongtao Ruan, Prof. Gokul Upadhyayula.

- Built a pipeline for direct mitotic-stage prediction from phase images of a fluorescence microscope, using ResNet and Swin Transformer models.
- Predicted mitotic stages via pseudo-labeling: reconstructed DNA structure from phase images with a GAN, then predicted stage from the reconstructed structure.
- Supported code deployment for non-CS lab members and taught introductory Python.

Efficient, High-quality Controllable Image Generation

Nov 2023 – May 2025

Lead. Advisors: Dr. Zhenliang He, Prof. Shiguang Shan.

- Surveyed diffusion and video-generation models; designed a controllable-generation scheme through extensive reproduction and local deployment.
- Proposed **CtrlLoRA-XL**, a cost-efficient, high-fidelity framework for multi-condition controllable image generation on SDXL, cutting GPU and training-image requirements by over 99% for adapting to novel conditions while preserving fidelity.
- Released the full implementation and UI as open source; drafted and submitted to IJCV.

Interactive Tenosynovitis Prevention via Deep Learning

Oct 2023 – Mar 2024

Lead. Advisors: Prof. Wei Sun, Prof. Feng Tian.

- Built a video dataset of tenosynovitis-relief gestures; used YOLOv5 to extract 21 hand keypoints, with data augmentation.
- Designed an LSTM gesture-recognition network (96.68% accuracy) and an interactive rehabilitation game; published as GIGT (ICCV 2024).

Few-shot PolSAR Image Classification with Diffusion Models

Jun 2023 – Sep 2023

Research Assistant. Advisor: Prof. Haixia Bi.

- Developed a self-supervised feature extractor based on DDPM and ViT, improving accuracy by 13.9% over CNNs and 3.6% over the previous SOTA.
- Ran comparative experiments on four A100 GPUs; co-authored the IGARSS 2024 paper.

Pipelined CPU Architecture (LoongArch / RISC-V)

Feb 2023 – Dec 2023

Researcher. Advisors: Prof. Wenxiang Wang, Prof. Weiwu Hu.

- Built a pipelined CPU on the RISC-V and LoongArch32 ISAs with data- and control-hazard resolution.
- Added exception handling, TLB caching, hardware deep-learning modules, and a DMA/AXI bus; ported the xv6 OS.

TEACHING & PROFESSIONAL EXPERIENCE

Teaching Assistant, Lower-Division Undergraduate Seminar, UCAS

Sep 2024 – Jan 2025

Advisor: Prof. Shuqiang Jiang. Supported research discussions and helped students scope initial research directions.

Summer Intern, ZhenTec Technology, Shaanxi, China

Jul 2022 – Aug 2022

Advisor: Dr. Haochong Wang. Helped develop lower-limb rehabilitation robots using brain-computer interfaces; designed interactive training games (e.g., racing, shooting) for self-guided rehabilitation.

SKILLS & TESTS

Programming: Python, C/C++, Verilog, Go, Java, LaTeX. **Tools:** Git/GitHub, Overleaf.

TOEFL: 103 (R28 / L29 / S23 / W23), 2024. **GRE:** 323 (V153 / Q170), 2023.